WEST BENGAL STATE COUNCIL OF TECHNICAL & VOCATIONAL EDUCATION AND SKILL DEVELOPMENT

(A Statutory Body under West Bengal Act XXVI of 2013)
(Technical Education Division)
Karigari Bhavan, 4th Floor, Plot No. B/7, Action Area-III, Newtown, Rajarhat, Kolkata–700 160

WBSCTVESD Curriculum for Diploma Courses in Engineering and Technology

Semester I
(Common to all Branches)

Sl. No	Category of Course	Course Title	Hours per week			Total contact hrs/ week	Credits	Marks
			L	T	P			
1.	Basic Science	Mathematics-I	2	1	0	3	3	100
2.	Basic Science	Applied Physics-I	2	1	0	3	3	100
3.	Basic Science	Applied Chemistry	2	1	0	3	3	100
4.	Humanities & Social Science	Communication Skills in English	2	0	0	2	2	100
5.	Engineering Science	Engineering Graphics	0	0	3	3	1.5	100
6.	Engineering Science	Engineering Workshop Practice	0	0	3	3	1.5	100
7.	Basic Science	Applied Physics-I Lab	0	0	2	2	1	100
8.	Basic Science	Applied Chemistry Lab	0	0	2	2	1	100
9.	Humanities & Social Science	Sports and Yoga	0	0	2	2	1	100
10.	Humanities & Social Science	Communication Skills in English Lab	0	0	2	2	1	100
	Total Credits and Marks						**18**	**1000**

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Syllabus for Engineering Mathematics-I

Course Title:	Engineering Mathematics-I
Course Code:	BS101/M-I
Number of Credits :	3(L: 2+1: T) P: 0
Pre Requisites :	1) Basic Formulae of Algebra, Trigonometry should be known 2) Preliminary knowledge of the vector
Total Contact Hours :	45 hrs.

Aim: Engineering Mathematics is the backbone of engineering students. The curriculum of mathematics has undergone changes from time to time in accordance with the need of engineering branches. The revised syllabus has been designed keeping in view the emerging needs of all categories of students. Great emphasis has been laid on the application of various contents like algebra, complex numbers, vectors, trigonometry and derivative. This course will develop analytical abilities to make exact calculations and provide a continuing educational base for the students.

Course Objectives: After the completion of the course the students will be able to

- apply the basic concepts of logarithm, complex number, quadratic equation and binomial theorem for solving the engineering and practical problems.
- find the solutions of vector oriented problems like work done, moment etc by applying vector algebra.
- simplify trigonometric expressions and solve trigonometric equations which will be useful in solving the scientific problems.
- analyze limit, continuity, derivatives of different functions and physical interpretation of derivatives which will be applicable in real situation.

Course Content

Unit-1

Algebra

1.1 Logarithm: 3 Hours

1.1.1 Definition of natural and common logarithm.

1.1.2 General Properties of logarithm and simple problems

1.2 Complex Numbers: 5 Hours

1.2.1 Definition of Complex numbers, Real and Imaginary parts of a complex number, Equality of two complex numbers, Conjugate of a complex number

1.2.2 Modulus and Argument of a complex number and simple problems

1.2.3 Polar and Cartesian forms of a complex number and their relation.

1.2.4 Algebraic operations (Addition, Subtraction, multiplication, Division) of complex numbers

1.2.5 De Moivre's Theorem (without proof) and simple problems.

1.2.6 Cube roots of unity and their properties with problems.

1.3 Quadratic Equations: 4 Hours

1.3.1 Definition of Quadratic Equations.

1.3.2 Finding the roots of a quadratic equation, conjugate roots& simple problems

1.3.3 Nature of the roots using discriminant & problems

1.3.4 Relation between roots and co-efficients & problems

1.3.5 Formation of quadratic equations if roots are given.

1.4 Binomial Theorem: 4 Hours

1.4.1 Definition of factorial of a number, permutation(n_{P_r})& combination (n_{C_r}) with formula only

1.4.2 Binomial Theorem (without proof) for any index, simple problems on positive index only

1.4.3 General Term and Middle Term and problems

1.4.4 Expansion of$(1+x)^{-1}, (1-x)^{-1}, where\ |x|<1$, exponential & logarithmic series only (no problem)

Unit-2

Vector Algebra 7 Hours

2.1 Definition of vector and types of vectors

2.2 Concept of a position vector and Ratio formula& simple problems

2.3 Rectangular resolution of a vector

2.4 Equality, addition, subtraction of vectors and multiplication of a vector by a scalar

2.5 Scalar (dot) and Vector (cross) product of two vectors with properties & simple problems

2.6 Application of dot product -- work done by a force, projection of a vector upon another

2.7 Application of cross product -- finding area of a triangle and parallelogram, moment of a force

Unit-3

Trigonometry 10 Hours

3.1 Concept of trigonometrical angles, measurement of angles in degree, radian and grade & their relation only.

3.2 Trigonometrical ratios of angles, associated angles, Trigonometric ratios of some standard angles, problems

3.3 Compound angles formula (without proof), multiple, sub-multiple angles & simple problems

3.4 Solutions of Trigonometrical Equations, simple problems (angle lies between 0 and 2π)

3.5 Inverse Circular Function & simple problems

3.6 Properties of triangle, basic formulae only

Unit-4

Function, Limit & Continuity, Derivative

4.1 **Function** 2 Hours

4.1.1 Definition of variables & constants

4.1.2 Definition of function with examples, domain and range of a function

4.1.3 Types of functions (even-odd, increasing-decreasing, inverse, periodic) with simple examples

4.1.4 Graph of trigonometric functions, sin x, cos x, tan x only

4.2 **Limit & Continuity** 2 Hours

4.2.1 Definition of limit (with left hand limit & right hand limit), Fundamental Theorem on limit (only statement), standard limits and simple problems

4.2.2 Continuity of functions, elementary test for continuity of functions (finite limit)

4.3 **Derivative** 8 Hours

4.3.1 Definition of derivatives

4.3.2 Derivatives of standard functions

4.3.3 Rules of differentiation of sum, difference, product and quotient of functions.

4.3.4 Derivatives of composite functions (Chain Rule)

4.3.5 Derivatives of inverse circular functions, implicit functions and logarithmic differentiation

4.3.6 Derivative of parametric functions, derivative of a function with respect to another function

4.3.7 Second order derivatives with simple problems

4.3.8 Application of derivatives –Physical & Geometrical interpretation of derivative, checking increasing-decreasing functions, finding velocity & acceleration, Maxima-Minima of function of single variable with simple problems.

Examination Scheme:

A. Semester Examination pattern of 60 marks:

1. Objective questions- 20 marks (1 mark each question), (At least 5 question from each **group**)
2. Subjective questions- 40 marks (at least 2 questions of 10 marks from each **group**)

- **Group- A** contains Unit-1 & Unit-2 (At least 40 marks); **Group-B** contains Unit-3 (At least 20 marks); **Group-C** contains Unit-4(At least 20 marks)

N.B.- Student will answer objective type questions of 20 marks and for subjective question of 40 marks, taking at least one question from each **group** of the above three **groups**.

B. For the internal Assessment 40 marks:

1. Class Test Examination/Internal Examination; 20 marks; choose best two out of three Class Test Examinations/ Internal Examinations
2. Class Attendance; 10 marks
3. Viva/ Quiz/Presentation/Assignment/Project/Report etc.; 10 marks

Text Books & Reference -

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, New Delhi
2. Engineering Mathematics, Konch & Dey, Bhagabati Publication
3. Engineering Mathematics, Babu Ram, Pearson
4. Trigonometry, S.L.Loney, S.Chand & Co.
5. Higher Algebra, H. S. Hall & Knight, Book Palace, New Delhi
6. Advanced Engineering Mathematics, E. Kreyszig, Wiley
7. Engineering Mathematics, A. Sarkar, Naba Publication
8. Diploma Engineering Mathematics, B. K. Paul, U.N.Dhar & Sons
9. Analytic Geometry Two & Three Dimensional and **Vector Analysis**, R. M. Khan, New Central Book Agency
10. Higher Algebra: Classical, S.K. Mapa, Sarat Book House
11. Introduction to Real Analysis, S.K. Mapa, Sarat Book House
12. Engineering Mathematics, Reena Garg, Khanna Publishing House, New Delhi
13. Calculus and Analytic Geometry, G. B. Thomas, R. L. Finney, Addison Wesley
14. Engineering Mathematics, V. Sundaram, R. Balasubramanian, K.A. Lakshminarayanan, Vikas Publishing House.
15. Advanced Engineering Mathematics, Reena Garg & Chandrika Prasad, Khanna Publishing House, New Delhi
16. Web portal: https://www.ndl.gov.in/homestudy/science
    https://ncertbooks.ncert.gov.in/login
    https://epathshala.nic.in//
    https://webscte.co.in/
    https://en.wikipedia.org/wiki/
    https://openlibrary.org/
    https://www.youtube.com/
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Applied Physics for Sem-I (Theory)

Reviewed and prepared by Syllabus-Sub-committee, on the basis of recommendation of AICTE

Sem –I (Theory)

Course Code	:	BS103
Course Title	:	Applied Physics-I
Number of credits	:	3 (L: 2, T: 1, P: 0)
Prerequisites	:	High School Level Physics
Course Category	:	BS

Course Content:

Unit 1: Physical world, Units and Measurements

Physical quantities; fundamental and derived, Units and systems of units (CGS and SI units),

Dimensions and dimensional formulae of physical quantities, Principle of homogeneity of dimensions, Dimensional equations and their applications (conversion from one system of units to other, checking of dimensional equations and derivation of simple equations), Limitations of dimensional analysis.

Measurements: Need, measuring instruments, least count, types of Measurement (direct, indirect), Errors in Measurements (systematic and random), absolute error, relative error, error propagation, error estimation and significant figures.

Unit 2: Force and Motion

Force, Momentum, Conservation of linear momentum, its applications such as recoil of gun, numerical problems rockets (concept only), Impulse and impulsive force.

Circular motion, definition of angular displacement, angular velocity, angular acceleration, frequency, time period, relation between linear and angular velocity, linear acceleration and angular acceleration (related numerical), concept of Centripetal and centrifugal forces with examples (No derivation, only formula) banking of roads and bending of cyclist,concept and formula and numerical problems.

Unit 3: Work, Power and Energy

Work: Concept and units, examples of zero work, positive and negative work

Friction: concept, types, laws of limiting friction, coefficient of friction, reducing friction and its engineering applications, Work done in moving an object on rough inclined plane.

Energy and its units, kinetic energy and potential energy, Conservation of mechanical energy for freely falling bodies(simple numerical problems), transformation of energy (examples only).

Power and its units, power and work relationship, calculation of power (numerical problems).

Unit 4: Rotational Motion

Translational and rotational motion with examples, Definition of torque and angular momentum and their relation, Conservation of angular momentum (quantitative) and its applications.

Moment of inertia and its physical significance, radius of gyration for rigid body, Theorems of parallel and perpendicular axes (statements only), Moment of inertia of rod, disc, ring and sphere (hollow and solid); (Formulae only). Simple numerical problems.

Unit 5: properties of Matter

Elasticity: definition of stress and strain, moduli of elasticity, Hooke's law, significance of stress-strain curve.

Surface tension: Concept, units, cohesiv**e and adhesive forces**, angle of contact, Capillary rise (formula only), applications of surface tension, effect of temperature and impurity on surface tension.

Viscosity and coefficient of viscosity: terminal velocity, Stoke's law and effect of temperature on viscosity.

Hydrodynamics: Fluid motion, stream line and turbulent flow, Reynol**d**'s number Equation of continuity, Bernoulli's Theorem (only formula and numericals) and its applications (mention name only).

Unit 6: Heat and Thermometry

Concept of heat and temperature, basic concepts of measurements of heat and temperature, modes of heats transfer (conduction, convection and radiation with examples), Co-efficient of thermal conductivity simple numerical problems.

Expansion of solids, liquids and gases, coefficient of linear, surface and cubical expansions of solids and relation amongst them, specific heats Cp & Cv of a gas and their relationship (Mention only).

References:

1. Text Book of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi
2. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi.
3. Concepts in Physics by HC Verma, Vol. I & II, Bharti Bhawan Ltd. New Delhi
4. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
5. Engineering Physics by DK Bhattacharya & Poonam Tandan; Oxford University Press, New Delhi
6. Comprehensive Practical Physics, Vol,I & II, JN Jaiswal, Laxmi Publications (P) Ltd., New Delhi
7. Practical Physics by C.L. Arora, S. Chand Publication.
8. Comprehensive Physics Vol,I & II.

Applied Physics for Sem-I (Lab)

Course Code	:	BS107
Course Tittle	:	Applied Physics-I Labs
Numbers of Credits	:	1 (L:0, T:0, P:2)
Prerequisites	:	NIL
Course Category	:	BS

Course Objectives:

Study of Applied Physics aims to give an understanding of physical world by observations and predictions. Concrete use of physical principles and analysis in various fields of engineering and technology is very prominence. The course aims to supplement the factual knowledge gained in the lecture by first hand manipulation of apparatus. This will develop scientific temper and help to apply the basic concepts and principles in solving engineering and technology based problems. In addition, students get necessary confidence in handling equipment and thus learn various skills in measurement.

List of Practical's/Activities(To perform minimum 8 practical's).

1. To measure the volume of the material of a given hollow cylinder, using a Vernier calipers.
2. To determine the area of cross section of a thin wire using a screw gauge.
3. To determine radius of curvature of a convex and a concave mirror/ surface using a spherometer.
4. To find the co-efficient of friction between wood and glass using a horizontal board.
5. To determine force constant of a spring using Hook's law.
6. To find the moment of inertia of a flywheel.
7. To find the viscosity of a given liquid (Glycerin) by Stoke's law
8. To find the co-efficient of linear expansion of the material of a rod.
9. To **verify** Boyle's law.
10. To determine the relative density of sand by using a sp. gr. Bottle.

Reference books:

1. Text books of Physics for Class-XI & XII (Part-I & II); N.C.E.R.T., Delhi.
2. Comprehensive Practical Physics, Vol-I & II, JN Laxmi Publications (P) Ltd.,
3. Practical Physics by C.L. Arora, S. Chand Publication.

Applied Chemistry

Course Code	:	BS105
Course Title	:	Applied Chemistry
Number of Credits	:	3 (L: 2, T: 1, P: 0)
Prerequisites	:	High School Level Chemistry
Course Category	:	BS

Course Objectives:

There are numerous number materials used in fabricating and manufacturing devices for the comfort of life. The selection, characterization and suitability assessment of natural raw materials essentially requires principles and concepts of Applied Chemistry for technicians. On successful completion of this course content will enable technicians to understand, ascertain and analyse and properties of natural raw materials require for producing economical and eco-friendly finished products.

- Solve various engineering problems applying the basic knowledge of atomic structure and chemical bonding.
- Use relevant water treatment method to solve domestic and industrial problems.
- Solve the engineering problems using knowledge of engineering materials and properties.
- Use relevant fuel and lubricants for domestic and industrial applications
- Solve the engineering problems using concept of Electrochemistry and corrosion.

Instruction on question setting:

- Question paper contains three groups A, B and C. Unit 1 and unit 2 are included in group A, unit 3 and unit 4 in group B, unit 5 in group C.
- 20 (twenty) number of questions are of objective types consisting of all groups, each carrying 1 (one) mark.
- 5 (five) questions are to be answered taking at least one from each group (each question carries 8 marks).

Course Content:

- **Unit 1: Atomic Structure, Chemical Bonding and Solutions**

  Rutherford model of atom, Bohr's theory (expression of energy and radius to be omitted), and hydrogen spectrum explanation based on Bohr's model of atom, Heisenberg uncertainty principle, Quantum numbers - orbital concept. Shapes of s, p and d orbitals Pauli's exclusion principle, Hund's rule of maximum multiplicity Aufbau rule, electronic configuration.

  Type of chemical bonding: ionic, covalent, metallic and hydrogen bonds. Example of each type. Hybridization, sp^3, sp^2, sp, example: $BeCl_2$, BF_3, CH_4, NH_3, H_2O; structure of diamond, graphite.
  Solution - idea of solute, solvent and solution, methods to express the concentration of solution-molarity (M = mole per liter), ppm, mass percentage, volume percentage and mole fraction.

- **Unit 2: Water**

  Graphical presentation of water distribution on Earth (pie or bar diagram). Classification of soft and hard water based on soap test, salts causing water hardness, unit of hardness and simple numerical on water hardness.

  Cause of poor lathering of soap in hard water, problems caused by the use of hard water in boiler (scale and sludge, foaming and priming, corrosion etc), and quantitative measurement of water hardness by ETDA method, total dissolved solids (TDS) alkalinity estimation.

  1) Water softening techniques - soda lime process, zeolite process and ion exchange process.
  2) Municipal water treatment (in brief only) - sedimentation, coagulation, filtration, sterilization.

  Water for human consumption for drinking and cooking purposes from any water sources and enlist Indian standard specification of drinking water (collect data and understand standards).

- **Unit 3: Engineering Materials**

  Natural occurrence of metals - minerals, ores of iron, aluminium and copper, gangue (matrix), flux, slag, metallurgy - brief account of general principles of metallurgy. Extraction of iron from haematite ore using blast furnace, aluminium from bauxite along with reactions, reactions during copper extraction. Alloys – definition, purposes of alloying, ferrous alloys and non-ferrous with suitable examples, properties and applications.

  General chemical composition, composition based applications (elementary idea only details omitted):

  Port land cement and hardening, Glasses Refractory and Composite materials.

  Polymers - monomer, homo and co polymers, degree of polymerization, simple reactions involved in preparation and their application of thermoplastics and thermosetting plastics (using PVC, PS, PTFE, nylon - 6, nylon - 66, Bakelite only), rubber and vulcanization of rubber.

- **Unit 4: Chemistry of Fuels and Lubricants**

  Definition of fuel and combustion of fuel, classification of fuels, calorific values (HCV and LCV), calculation of HCV and LCV using Dulong's formula.

  Proximate analysis and ultimate analysis of coal solid fuel

  petrol and diesel - fuel rating (octane and cetane numbers),

  Chemical composition, calorific values and applications of LPG, CNG, water gas, coal gas, producer gas and biogas.

  Lubrication - function and characteristic properties of good lubricant, classification with examples, lubrication mechanism - hydrodynamic and boundary lubrication, physical properties (viscosity and viscosity index, oiliness, flash and fire point, could and pour point only) and chemical properties (coke number, total acid number saponification value) of lubricants.

- **Unit 5: Electro Chemistry**

  Electronic concept of oxidation, reduction and redox reactions.

  Definition of terms: electrolytes, non-electrolytes with suitable examples, Faradays laws of electrolysis and simple numerical problems.

  Elementary concept of pH and buffer.

  Industrial Application of Electrolysis –
  - Electrometallurgy
  - Electroplating
  - Electrolytic refining.

  Application of redox reactions in electrochemical cells –
  - Primary cells - dry cell,
  - Secondary cell - commercially used lead storage battery, fuel and Solar cells.

  Introduction to Corrosion of metals –
  - definition, types of corrosion (chemical and electrochemical), H_2 liberation and O_2 absorption mechanism of electrochemical corrosion, factors affecting rate of corrosion.

  Internal corrosion preventive measures –
  - Purification, alloying and heat treatment and

  External corrosion preventive measures: a) metal (anodic, cathodic) coatings, b) organic inhibitors.

Suggested Sessional work:

- **Unit 1: Atomic Structure, Chemical Bonding and Solutions**

  Assignments: Writing electronic configuration of elements up to atomic number 30 (Z= 30). Numerical on molarity, ppm, mass percentage, volume percentage and mole fraction of given solution.

  Seminar: 1. Quantum numbers,
  2. Discuss the metallic properties such as malleability, ductility, hardness, high melting point, conductance of heat and electricity, magnetic properties of metals.

  Projects: Model of molecules $BeCl_2$, BF_3, CH_4, NH_3, H_2O

- **Unit 2: Water**

  Assignments: Simple problems on hardness calculation.

  Seminar: 1. Quality and quantity requirement of water in house and industry.
  2. Quality of control measures of effluents (BOD & COD).

  Projects: Collect water samples from different water sources and measure of hardness of water.

- **Unit 3: Engineering Materials**

  Assignments: Preparation of table showing different ores of iron, copper and aluminium metals along with their chemical compositions and classify in to oxide sulphide halide ores.

  Seminar: Discuss the chemical reactions taking place in Blast Furnace in extraction of iron; Reactions occurring during extraction of copper and aluminium metals.

  Projects: Make table showing place of availability of different ores in India and show places on India map.

- **Unit 4: Chemistry of Fuels and Lubricants**

  Assignments: Calculation of HCV and LCV of fuel using fuel composition in Dulong's formula.

  Seminar: Chemical structure of fuel components influence on fuel rating.

  Projects: Mapping of energy recourses in India. Collection of data of various lubricants available in the market.

- **Unit 5: Electro Chemistry**

  Assignments: Simple problems on Faradays laws of electrolysis.

  Seminar: 1. Corrosion rate and units.
  2. Corrosion preventions.

  Projects: Mapping of area in India prone to corrosion. Collection of data of various electrochemical cells batteries used in equipment and devices and available in market. Visit to sites such as Railway station to watch corrosion area in railways and research establishment in and around the institution.

References/Suggested Learning Resources:

(a) Books:

1) Text Book of Chemistry for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi, 2017-18.
2) Agarwal, & Shikha, Engineering Chemistry, Cambridge University Press; New Delhi, 2015.
3) C.N. R. Rao, Understanding Chemistry, Universities Press (India) Pvt. Ltd., 2011.
4) S. S. Dara & S.S.Umare, Engineering Chemistry, S. Chand. Publication, New Delhi, New Delhi, 2015.
5) Jain & Jain, Engineering Chemistry, Dhanpat Rai and Sons; New Delhi, 2015.
6) S. Vairam, Engineering Chemistry, Wiley India Pvt.Ltd., New Delhi, 2013.
7) G. H. Hugar & A. N. Pathak, Applied Chemistry Laboratory Practices, Vol. l and Vol. II, NITTTR, Chandigarh, Publications, 2013-14.
8) Rajesh Agnihotri, Chemistry for Engineers, Wiley India Pvt.Ltd., 2014.

(b) Open source software and website address:

1) www.chemguide.co.uk/atommenu.html (Atomic structure and chemical bonding)
2) www.visionlearning.com (Atomic structure and chemical bonding)
3) wwwcheml.com (Atomic structure and chemical bonding)
4) https://www.wastewaterelearning.com/elearning/ (Water Treatment)
5) www.capital-refractories.com (Metals, Alloys, Cement, and Refractory Materials)
6) www.em-ea.org/guide%20books/book-2/2.1%20fuels%20and%20combustion.pdf (Fuel and Combustion)
7) www.chemcollective.org (Metals, Alloys)
8) www.wqa.org(Water Treatment)

Applied Chemistry Lab

Course Code	:	BS109
Course Title	:	Applied Chemistry Lab
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	BS

Course Objectives:

There are numerous number of materials used in fabricating and manufacturing devices for the comfort of life. The selection, characterization and suitability assessment of natural raw materials essentially requires principles and concepts of Applied Chemistry for technicians. The course aims to supplement the factual knowledge gained in the lectures by first hand manipulation of processes and apparatus. This will develop scientific temper and help to apply the basic concepts and principles in solving engineering problems.

LIST OF PRACTICALS:

Perform any 12 (twelve) Laboratory Practicals:

Volumetric and Gravimetric analysis.

1. Preparation of standard oxalic acid and standard potassium dichromate solution.
2. To determine strength of given sodium hydroxide solution by titrating against standard oxalic acid solution and phenolphthalein as indicator.
3. Standardization of potassium permanganate solution using standard oxalic acid and determination of percentage of iron present in given Hematite ore by $KMnO_4$ solution
4. a) Standardization of sodium thiosulphate using standard potassium dichromate solution by IODOMETRY.

b) Iodometric estimation of copper in copper pyrite ore.

5. Volumetric estimation of total acid number (TAN) of given oil.
6. Volumetric estimation of
    a. Total hardness of given water sample using standard EDTA solution.
    b. Alkalinity of given water sample using 0.01N sulphuric acid.
7. Proximate analysis of coal
    a. Gravimetric estimation moisture in given coal sample.
    b. Gravimetric estimation ash in given coal sample

Instrumental analysis

8. Determine the conductivity of given water sample.
9. Determination of the Iron content in given cement sample using colorimeter.
10. Determination of viscosity of lubricating oil using Redwood viscometer.
11. Determination of flash and fire point of lubricating oil using Able's flash point apparatus.
12. To verify the first law of electrolysis of copper sulfate using copper electrode.
13. Construction and measurement of emf of electrochemical cell (Daniel cell).
14. To study the effect of dissimilar metal combination on cell emf.
15. To apply thin layer chromatography for separation of mixture of inorganic/organic compounds.
16. Qualitative detection of ARSENIC in a given sample of water (~5 ppm solution of sodium arsenite)
17. Determination of dissolved oxygen in a sample of water.
18. Determination of pH value of unknown solution.

Reference book

VOGELS INORGANIC QUANTITATIVE ANALYSIS.

Members present in Syllabus Committee:

1. **Dr. Ujjval Kumar Bhattacharyya** (convener, lecturer, North Calcutta Polytechnic).
2. **Dr. Gandhi Kumar Kar** (Professor and head of the Dept. of Chemistry, Presidency University, Kolkata)
3. **Dr. Shyamal Kumar Chattopadhyay** (Professor IIEST, Shibpur).
4. **Dr. Mrinal Kanti Bain**, lecturer, Calcutta Technical School.
5. **Dr. Dipankar Maity**, lecturer, Birla Institute of Technology.
6. **Dr. Ranjal Paul**, Sr. Manager, Sun Pharmaceuticals Baroda.
7. **Prolay Roy,** lecturer, Memari Government Polytechnic.

Syllabus for Engineering Graphics

Name of the Course :	**Engineering Graphics**
Course Code : ES 101	Semester : First
Duration : 17 Weeks	Maximum Marks : : 100
Lecture : Nil	Assignment : 20
	Class Performance : 20
Tutorial : Nil	Viva Voce : 10
	Attendance : 10
Practical : 3 hrs./week	External (Viva, Sketch Book & Drawing Sheet) : 40
Credit : 1.5	

Aim :

1. The Course is aimed to develop the basic graphic skills so that the students can prepare Engineering Drawing in their practical field.
2. Understand the fundamentals of Engineering Graphics.
3. Read and interpret object drawing.
4. To develop the skills of Computer Aided Drafting and can easily cope up the skill of drafting for modern drawing offices/industries.

Objectives :

1. To develop sense of Scale and drawing technique of different curves and their application.
2. To develop drafting and sketching skills, to know the applications of drawing equipment and get familiarize with Indian Standards related to engineering drawing.
3. To develop concept of Orthographic Projection and to draw Orthographic Views for different objects.
4. To visualize three dimensional objects from Orthographic Views and to draw isometric views/projections.
5. To be familiar with AutoCAD and to develop the skill of drafting in AutoCAD by using different commands.

Pre-requisites :

1. Unambiguous and clear visualization.
2. Sound Pictorial Intelligence.

Unit No.	Contents
Unit 1	1.1 Letters and numerals (Single Stroke Vertical) 1.2 Conventions of lines and their applications 1.3 Concept of Representative Fraction (R.F), Reduced scale, Enlarged scale & Full Scale, Engineering Scale such as Plain Scale & Diagonal Scale. 1.4 Dimensional Techniques - Unidirectional System and Aligned System.
Unit 2	2.1 Geometrical Construction : a)Draw Regular Polygons, Ellipse, Parabola, hyperbola b) Draw Curve passing through given no. of points, cycloid, involute of a circle and polygon
Unit 3	3.1 Introduction to Orthographic Projection : Concept of First Angle & Third Angle of projection. Projections of lines(limited to both ends in 1st quadrant) : parallel to the reference planes, inclined to the reference planes (1st Angle Method) 3.2 Projections of solid body: Regular Polygonal Pyramid, Cylinder, Cone - inclined to only one reference plane (1st Angle Method)

Unit 4	4.1 Conversion of Pictorial Views into orthographic views (Simple Objects &First Angle Projection Method Only)
Unit 5	5.1 Introduction to Isometric Projections, Concept of Isometric Scale and Isometric Views 5.2 Conversion of Orthographic views into Isometric Views/Projections : Simple Objects e.g. regular prism, pyramid, cone, cylinder.
Unit 6	6.1 Free hand sketches of engineering elements such as thread profile, nuts, bolts, studs, set screws, washers, locking arrangements etc. and their conventional representation (For branches other than Mechanical Engineering, the teachers should select branch specific elements for free hand sketching)
Unit 7	7.1 Introduction to different commands and toolbars of AutoCAD a) Draw command : Lines, Circle, Polygon, Arc, Ellipse, Polyline, Fillet, Chamfer, Hatch, Array (Rectangular & Polar) etc. b) Modify Command : Offset, Trim, Extend, Erase, Fillet, Chamfer, Break, Lengthen, Copy, Move, Mirror, Stretch, Match Properties, Pedit etc. c) View : Zoom All, Zoom Window, Zoom Extent etc. d) Dimensioning : Dimension Setting, Linear, Aligned, Radial, Diameter, Leader, Angular etc. c) Text : Text Style, Dtext, Mtext, DDedit etc. d) Format : Limits, Layers, Pan etc. 7.2 Making a few simple 2D drawings in AutoCAD.

Sl. No.	Practical Exercise	Unit No.	Hrs.
1	Draw horizontal, vertical, 30˚, 45˚, 60˚, 75˚, different types of lines, dimensioning styles using Tee and Set Squares / Drafters. (Do this exercise in Sketch Book)	1	02
2	Write single stroke vertical alphabets and numerical (7:4 ratio) (Do this exercise in Sketch Book)	1	02
3	Draw regular geometric constructions and redraw/copy the given figure (Do this exercise in Sketch Book)	1	02
4	Draw at least two problems on plain scale and two problems on diagonal scales (Do this exercise in Sketch Book)	1	03
5	Draw problems on ellipse, parabola, involute and cycloid (Do this exercise in Sketch Book)	2	06
6	Draw at least two problems on lines and two problems on solid (First Angle Projection method) (Do this exercise in A2 size drawing sheet)	3	09
7	Plan, Elevation and Side View of at least two pictorial views to be drawn on one A2 size Drawing Sheet along with dimensions	4	06
8	At least Two Isometric Views and two Isometric Projections to be drawn on one A2 size Drawing Sheet	5	06

9	Draw free hand sketches / conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washers, locking arrangements etc.	6	03
10	Simple geometrical figures such as triangle, rectangle, polygon, circle, ellipse, and simple orthographic views of brackets, gaskets etc. to be drawn in AutoCAD and Printout to be taken on A4 size Sheet. At least two sheets to be prepared.	7	12
	Total Period		51

Text Books:

Name of Authors	Title of the Book	Edition	Name of the Publisher
N.D.Bhatt	Engineering Drawing	53rd Edition	Charotkar Publishing House
R.K.Dhawan	Engineering Drawing	Revised Edition 2015	S.Chand & Co.
K. Venugopal & V. Prabhuraja Raja	Engineering Drawing and Graphics + AutoCAD		New Age Publication

Reference Books:

Name of Authors	Title of the Book	Edition	Name of the Publisher
S.N. Lal	Engineering Drawing		CENGAGE
Basant Agrawal & C M Agrwal	Engineering Drawing		Tata McGrew Hill
Dhananjay A Jolhe	Engineering Drawing with an Introduction to AutoCAD		Tata McGrew Hill
S.P.Dey	AutoCAD 2014		Vikas Publisher
P.S. Gill	Engineering Drawing		S.K.Kataria & Sons.

References :

1.Engineering Graphics Syllabus of AICTE Model Curriculum 2019.

2. Syllabus of Technical Drawing (Old) of W.B.S.C.T.E

Engineering Workshop Practice

Course Code	:	ES 112
Course Title	:	Engineering Workshop Practice
Number of Credits	:	3(L-0, T-0, P-3)
Prerequisites	:	
Course Category	:	Engineering Science
Semester	:	First

COMPETENCY

The Engineering Workshop Practice Course should be taught and implemented with the aim to develop the following Course Outcomes (COs) so that the students will be able to prepare simple jobs on the shop floor of the engineering workshop.

COURSE OUTCOMES

The theory and practical experiences associated with the course, the students will gain the following industry oriented COs:

- Read and interpret job drawings
- Select tools, equipment and machineries according to the job.
- Use the hand tools in different shops for performing different operations
- Prepare the job according to drawing
- Adopt safe working practice.
- Maintenance of workshop tools, equipment and machineries.
- Acquaint with the specifications on all raw materials, tools and equipments used.

COURSE CONTENT

Course Content: There are 9 (nine) modules out of which 6 modules should be taught. Selection of these six modules should be in the following manner.
Module 1, 2 and 3 are compulsory and any three from rest as deemed fit for the branch and availability in the institute.

Module No. & Name	Details of Workshop Content	Hours
1. Electrical Wiring	**1. Introductory Session** • Safety precautions to be followed in Electrical Works • Electric shock, methods of shock treatment • Safety measure: Earthing, Fuse, circuit breakers, etc. • Different types of wire-gauge & strands and their applications • Different tools used in Electrical wiring works • General wiring accessories & their uses.	

	• Types of switches and their application. • Types of wiring and joints. **Practical (any three)- Group works** 1. Study/ demonstration of single phase electrical service connection from pole to house including consumer installation. 2. Study of different types of wiring and connection of Single Phase Motor (Fan). 3. Circuits for one lamp controlled by one switch by surface conduit wiring, 4. Lamp circuits- connection of lamp and socket by separate switches, Connection of Fluorescent lamp/tube light, 5. Simple lamp circuits- install bedroom lighting 6. Simple lamp circuits- install stair case wiring. 7. Wiring of calling-bell	10 hrs
2. Fitting	**1. Introductory Session** • Demonstration of different tools and equipment used in fitting shop. • Study of different precious measuring instrument such as micrometer, vernier calipers, bevel protectors. • Care and maintenance of the above mentioned tools and equipment. • Study of drilling machines and power tools used in fitting shop. • Safety measure to be followed in Fitting shop. **2. Practical/ Demonstration** • Demonstration of different fitting job operations like chipping, filing, drilling, tapping, sawing, cutting etc. **Job Preparation -Individual Works** • One simple fitting job involving practice of chipping, filing, marking, drilling, tapping, cuttingetc.	10 hrs
3. Welding	**1. Introductory Session** • Purpose of welding, advantages & disadvantages of it over other joining processes. • Types of welding processes (in brief) • Specification, usage, care & maintenance of various welding machines, tools & equipment used in the shop. • Selection of welding methods and electrodes. • Safety measures & equipment required while working in welding shop.	10 hrs

	2. Practical/ Demonstration • How to start an Arc & use it for Arc Welding, sustainably. • Demonstration of various welding m/c, tools, equipment available in the shop. • Demonstration of shielded metal arc welding (SMAW), Gas welding and cutting. • Repairing of broken metal structures using welding. • Defects of welding & their remedies. Job Preparation (Any One)-Individual Works Job 1 – Butt Joint Job 2 – Lap Joint Job 3 – 'T' Joint	
4. Carpentry	**1. Introductory Session** • Raw materials used in carpentry shop: wood & alternative materials. • Specification, usage, care & maintenance of various tools, equipment and machineries used in the Carpentry shop. • Types of wood. Difference between hard and soft wood. • Timber: characteristics, usage and defects. Difference between wood and timber. • Seasoning of wood. • Different types of joints such as cross half-lap joint, through tenon and mortise joint, dove tail joints, etc. • Safety measures to be taken in carpentry shop. **2. Practical/Demonstration** • Demonstration of use of different tools, equipment and machineries. • Demonstration of different wood working processes, like plaining, marking, chiseling, grooving, turning of wood etc. Job Preparation -Individual Works One simple job involving any one joint.	10 hrs
5. Sheet Metal Working	**1. Introductory Session** • Briefing on different types of sheet metal, like Stainless Steel Sheet Metal, Copper Sheet Metal, Brass Sheet Metal, Corrugated Sheet Metal, Galvanized Sheet Metals etc., and their uses. • Demonstration of different types of Tools& machines and their use in sheet metal work.	

	• Demonstration of different types sheet metal joints and their applications. • Demonstration of different types of sheet metal working: cold working, hot working, warm working, bending, drawing, end curling, shearing, piercing, sheet metal presses, etc. • Sheet metal joining operation like welding, brazing, soldering and riveting. • Safety measure to be followed in sheet metal work. **2. Practical:** Making of any simple job(example: sheet metal mug) involving different sheet petal operations including soldering and riveting.	10 hrs
6. Smithy/Forging	**1. Introductory Session** • Purpose of Smithy / Forging Works • Different types of Hearths used for Smithy / Forging works • Specification, usage, care and maintenance of various tools and equipment used in the shop. • Types of raw materials used in Smithy / Forging shop & their required temperature for it. • Types of fuel used in hearth and the respective maximum temperature obtained. • Uses of Fire Bricks & Clays in Smithy/Forging Work Shop. • Types of heat treatment processes involved in Smithy / Forging shop and its effect on forged items. • Hot forge & cold forge utility. • Safety measures & equipment required while working in Smithy / Forging Shop **2. Practical/ Demonstration** • Practice / Demonstration of firing of hearth / Furnace, Cleaning of Clinkers and Temperature Control of Fire. • Practice / Demonstration on different basic Smithy / Forging operations such as Upsetting, Drawing down, Setting down, Necking, Cutting, Bending, Fullering, Swaging, Punching and Drifting etc. Demonstration on making of • Cube, hexagonal cube, hexagonal bar from round bar. • Hexagonal /octagonal flat chisel including tempering of edges.	10 hrs

	Job Preparation (Any one) – **group effort** by students • Job 1 Making a cold / hot flat chisel • Job 2 Simple Tong • Job 3 Production of any other utility tools/ items -e.g. Chain-links, door ring, hexagonal bolt / square shank boring tool, fan hook (long S-type) etc.	
7. Machine shop Practice	1. **Introductory Session** • Purpose of a machine shop/ workshop in industry. • Demonstration of all available tools & tackles of machine shop. • Inculcation of basic idea of machine tool; differentiation between machine & machine tool. • Familiarization of all machine tools in the machine shop. • Safety precautions for working in a machine shop. 2. **Practice in Machine Shop** • Identification of different parts of a lathe and utility of those parts. • Demonstration of all possible machining operation on a lathe, e.g. turning, facing, parting, taper turning, drilling, threading etc. • Demonstration to operate a drilling machine or shaping machine and identifying different parts of that m/c tool. Job Practice-(Individual work) Preparation of one job in Lathe, involving simple machining operations (e.g. turning, facing, grooving, threading, knurling etc.).	10 hrs
8. Electronic Shop	**1. Introductory Session** • Discussion on active and passive electronic components. • Discussion on soldering and its use. • Introduction on multi-meter and its use. • Discussion on use of test equipment in fault finding. • Discussion on resistor, capacitor, amplifiers, relay, diodes, zener diode and LEDs. • Safety measure to be followed in electronic shop. **2. Practical/ Demonstration** • Familiarization, identification and testing of active and passive components. • Soldering and de-soldering practice.	10 hrs

	• Use of Multi meter (both Analog and digital). • Demonstration of resistor, capacitor, amplifiers, relay, diodes, zener diode and LEDs.	
9. Demonstration	Following demonstration may be conducted: 1. Demonstration of measurement of Current, Voltage, Power and Energy for an electrical system/ wiring. 2. Demonstration of advanced power tools such as pneumatic tools, electrical portable grinding tools and accessories. 3. Demonstration of bourdon tube pressure gauge. 4. Demonstration of ball bearing and roller bearing. 5. Demonstration of portable power tools for Cutting and drilling, etc.	10 hrs

LEARNING OUTCOMES (LOs)

At the end of the course, the student will be able to:

- Understand the basic safety measure to be followed in different works.
- Understand basic engineering processes for manufacturing and assembly.
- Understand and interpret job drawings, produce jobs, and inspect the job for specified dimensions.
- Understand the various types of wiring systems and acquire skills in electrical wiring.

References

[1] S.K. HazraChaudhary, Workshop Technology, Volume I&II, Media Promoters and Publishers, Mumbai.

[2] B.S. Raghuwanshi, Workshop Technology, Volume I&II, DhanpathRai and Sons, New Delhi 2014

[3] K. Venkat Reddy, Workshop Practice Manual, BS Publications, Hyderabad 2014

[4] Kents Mechanical Engineering Hand book, John Wiley and Sons, New York

[5] H.S. Bawa, Workshop Practice, Mcgrawhill HED

[6] R.P. Singh, Electrical Workshop: Safety, Commissioning, Maintenance & Testing of Electrical Equipment, Wiley

COMMUNICATION SKILLS IN ENGLISH

Course Code	HS101
Course Title	Communication Skills in English
Number of Credits and L-T-P	2 [L – 2, T – 0, P - 0]
Course Category	Humanities & Social Sciences

Course Objectives:

The course intends to:

- Impart the basic theory and practice of communication.
- Make students aware about the importance of soft skills as essential Life skills for both personal development and professional excellence.
- Train students for improving their reading, writing, listening and speaking skills.
- Clear the learners' misconceptions and misunderstandings of English grammar.
- Give students the skills and the confidence required to speak English fluently.

Course Contents

Unit1: Communication: Theory and Practice: (Marks division: 04)

- Basic of Communication: Introduction, meaning and definition, process of communication etc.
- Types of communication: formal and informal, verbal, non-verbal and written. Barriers to effective communication.
- 7 Cs for effective communication (considerate, concrete, concise, clear, complete, correct, courteous).
- Technical Communication.

Unit-2 Soft Skills for Professional Excellence: (Marks division: 08)

- Introduction: Soft Skills and Hard Skills.
- Time Management.
- Motivation
- Stress Management.
- Emotional Intelligence.
- Self-awareness.
- Problem solving skills
- Decision Making
- Interpersonal Skills

Unit-3: Reading Comprehension: (Marks division: 10)

- Comprehension of a written text

- Note Taking.

Unit-4: Professional Writing:

- Writing Reports - (Marks division: 20)
- Writing Emails – (Marks division: 10)
- Writing Memo – (Marks division: 10)
- Job Application Letters – (Marks division: with CV - 15)
- CV/Resume – (Marks division: 10)

Unit-5: Vocabulary and Grammar: (Marks division: 08)

- Remedial Grammar and Exercises
- Parts of speech, active and passive voice, tenses etc.

Course Outcomes:

After the completion of the Course, the learners will be able to:

- Understand, remember and effectively apply the Theory of Communication.
- Inculcate the essential soft skills in their life.
- Substantially improve their reading, writing, listening and speaking skills.
- Clarify and/ or remedy their misunderstanding and misconceptions about grammar which they have learnt till date.
- Overcome their hesitations, inhibitions and fear and gain confidence in communicating in simple, easily understandable English.

#Note: *No Subjective questions are to be set for the Theoretical examinations from Units 1, 2 and 5 of the syllabus.*

COMMUNICATION SKILLS IN ENGLISH - LAB

Course Code	HS105
Course Title	Communication Skills in English Lab
Number of Credits and L-T-P	2 [L - 0, T – 0, P - 2]
Course Category	Humanities & Social Sciences

Course Objectives:

The course intends to:

- Impart the basic theory and practice of communication.
- Make students aware about the importance of soft skills as essential Life skills for both personal development and professional excellence.
- Train students for improving their reading, writing, listening and speaking skills.

- Clear the learners' misconceptions and misunderstandings of English grammar.
- Give students the skills and the confidence required to speak English fluently.

Sessional 100 Marks (Internal: 60 Marks + External: 40 Marks)

Course Contents:

- Introducing yourself and others:
    - Talking about yourself
    - Describing people

- Speaking about your free time:
    - Talking about your free time
    - Giving reasons for things you like
    - Discussing daily routines

- Giving instructions and advice:
    - Giving and following instructions
    - Giving route directions
    - Advising about places/ tourist spots etc.

- Out and about:
  Talking about
    - Going shopping
    - Eating in a restaurant
    - Films and Television
    - Holidays
    - Social Events

- Speaking and listening:
    - College Life
    - Talking about the World, Weather, Environment, etc.

- Enhancing employability and professional skills:
    - Job Interviews
    - Group discussions
    - Professional Presentations

Course Outcomes:

After the completion of the Course, the learners will be able to:

- Understand, remember and effectively apply the Theory of Communication.
- Inculcate the essential soft skills in their life.
- Substantially improve their reading, writing, listening and speaking skills.

- Clarify and/ or remedy their misunderstanding and misconceptions about grammar which they have learnt till date.
- Overcome their hesitations, inhibitions and fear and gain confidence in communicating in simple, easily understandable English.

Sessional Activities

The students should prepare and write down the following activities in their sessional file and submit it to the Internal Examiner. The signed sessional file should be brought during the External Sessional viva examination.

1. A brief introduction about themselves.
2. A brief write-up about how they spend their leisure time.
3. Write down at least five conversations in dialogue from topics listed in the sessional syllabus (students should practice all types of conversation given in the list).
4. (i) List the Dos and Don'ts of a job interview.
   (ii) Write down anticipated questions (HR Round) asked during a job interview and write down appropriate answers to them.
5. List the Dos and Don'ts of an effective Group Discussion.
6. Prepare a professional Powerpoint Presentation (individually/ in groups) on a topic. The printouts of the slides should be attached in the sessional file.

Reference Books

01 English Skills for Technical Students (West Bengal State Council of Technical Education in collaboration with the British Council), Orient Black Swan
02 Communication Skills by Shri Sanjay Kumar & Ms. Pushp Lata, Oxford University Press
03 Improve Your Writing by Shri U.N. Arora & Ms. Lakshmi Chandra, Oxford University Press
04 Listening Skills by Shri Ian Mackay, Universities Press
05 Improving Teaching & Learning of English & Communication Skills in West Bengal Polytechnics Student Book, British Council
06 Development of Life Skills – II by Dr. Shalini Verma, Vikas Publishing House Pvt. Ltd.
07 Business Correspondence and Report Writing: A Practical Approach to Business and Technical Communication by R. C. Sharma, Krishna Mohan, Virendra Singh Nirban
08 Effective Technical Communication by Ashraf Rijvi
09 Personality Development and Soft Skills by Barun K. Mitra, Oxford University Press

Sports and Yoga

Course Code	HS103
Course Title	Sports and Yoga
Number of Credits and L-T-P	1 [L – 0, T – 0, P - 1]
Course Category	Humanities & Social Sciences
Prerequisites	Nil

Course Objectives:

- To make the students understand the importance of sound health and fitness principles as they relate to better health.
- To expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness.
- To create a safe, progressive, methodical and efficient activity based plan to enhance improvement and minimize risk of injury.
- To develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Course Content:

Introduction to Physical Education

- Meaning & definition of Physical Education
- Aims & Objectives of Physical Education
- Changing trends in Physical Education

Olympic Movement

- Ancient & Modern Olympics (Summer & Winter)
- Olympic Symbols, Ideals, Objectives & Values
- Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc.)

Physical Fitness, Wellness & Lifestyle

- Meaning & Importance of Physical Fitness & Wellness
- Components of Physical fitness
- Components of Health related fitness
- Components of wellness
- Preventing Health Threats through Lifestyle Change

- Concept of Positive Lifestyle

Fundamentals of Anatomy & Physiology in Physical Education, Sports and Yoga

- Define Anatomy, Physiology & Its Importance
- Effect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.)

Kinesiology, Biomechanics & Sports

- Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports
- Newton's Law of Motion & its application in sports.
- Friction and its effects in Sports.

Postures

- Meaning and Concept of Postures.
- Causes of Bad Posture.
- Advantages & disadvantages of weight training.
- Concept & advantages of Correct Posture.
- Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, BowLegs and Scoliosis.
- Corrective Measures for Postural Deformities

Yoga

- Meaning & Importance of Yoga
- Elements of Yoga
- Introduction - Asanas, Pranayama, Meditation & Yogic Kriyas
- Yoga for concentration & related Asanas (Sukhasana; Tadasana; Padmasana & Sha-shankasana)
- Relaxation Techniques for improving concentration - Yog-nidra

Yoga & Lifestyle

- Asanas as preventive measures.
- Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana.
- Obesity: Procedure, Benefits & contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana.
- Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana.

- Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana.
- Asthema: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.

Training and Planning in Sports

- Meaning of Training
- Warming up and limbering down
- Skill, Technique & Style
- Meaning and Objectives of Planning.
- Tournament – Knock-Out, League/Round Robin & Combination.

Psychology & Sports

- Definition & Importance of Psychology in Physical Edu. & Sports
- Define & Differentiate Between Growth & Development
- Adolescent Problems & Their Management
- Emotion: Concept, Type & Controlling of emotions
- Meaning, Concept & Types of Aggressions in Sports.
- Psychological benefits of exercise.
- Anxiety & Fear and its effects on Sports Performance.
- Motivation, its type & techniques.
- Understanding Stress & Coping Strategies.

Doping

- Meaning and Concept of Doping
- Prohibited Substances & Methods
- Side Effects of Prohibited Substances

Sports Medicine

- First Aid – Definition, Aims & Objectives.
- Sports injuries: Classification, Causes & Prevention.
- Management of Injuries: Soft Tissue Injuries and Bone & Joint Injuries

Sports/ Games

Following sub topics related to any one Game/Sport of choice of student out of: Athletics, Badminton, Basketball, Chess, Cricket, Kabaddi, Lawn Tennis, Swimming, Table Tennis, Volleyball, Yoga etc.

- History of the Game/Sport.
- Latest General Rules of the Game/Sport.
- Specifications of Play Fields and Related Sports Equipment.
- Important Tournaments and Venues.
- Sports Personalities.
- Proper Sports Gear and its Importance.

References:

1. Modern Trends and Physical Education by Prof. Ajmer Singh.
2. Light On Yoga By B.K.S. Iyengar.
3. Health and Physical Education – NCERT (11th and 12th Classes)

Course Outcomes:

On successful completion of the course the students will be able to:

(i) Practice Physical activities and Hatha Yoga focusing on yoga for strength, flexibility, and relaxation.
(ii) Learn techniques for increasing concentration and decreasing anxiety which leads to stronger academic performance.
(iii) Learn breathing exercises and healthy fitness activities
(iv) Understand basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination.
(v) Perform yoga movements in various combination and forms.
(vi) Assess current personal fitness levels.
(vii) Identify opportunities for participation in yoga and sports activities.
(viii) Develop understanding of health-related fitness components: cardio respiratory endurance, flexibility and body composition etc.
(ix) Improve personal fitness through participation in sports and yogic activities.
(x) Develop understanding of psychological problems associated with the age and lifestyle.
(xi) Demonstrate an understanding of sound nutritional practices as related to health and physical performance.
(xii) Assess yoga activities in terms of fitness value.
(xiii) Identify and apply injury prevention principles related to yoga and physical fitness activities.
(xiv) Understand and correctly apply biomechanical and physiological principles elated to exercise and training.

WEST BENGAL STATE COUNCIL OF TECHNICAL & VOCATIONAL EDUCATION AND SKILL DEVELOPMENT

(A Statutory Body under West Bengal Act XXVI of 2013)

(Technical Education Division)

Karigari Bhavan, 4th Floor, Plot No. B/7, Action Area-III, Newtown, Rajarhat, Kolkata–700 160

WBSCTVESD Curriculum for Diploma Courses in Engineering and Technology

Sl. No.	Category of Course	Course Title	Hours per week			Total contact hrs/ week	Credits	Marks	Examination Scheme				
			L	T	P								
	Theory Subjects								External Assessment 60		Internal Assessment-40		
									End Sem. Examination	Mid Sem. Test	Quizzes, viva-voce, Assignment	Class Attendance	
1.	Basic Science	Mathematics-II	3	1	0	4	4	100	60	20	10	10	
2.	Basic Science	Applied Physics-II	2	1	0	3	3	100	60	20	10	10	
3.	Engineering Science	Introduction to IT Systems	2	0	0	2	2	100	60	20	10	10	
4.	Engineering Science	Fundamentals of Electrical & Electronics Engineering	2	1	0	3	3	100	60	20	10	10	
5.	Engineering Science	Engineering Mechanics	2	1	0	3	3	100					
	Practical Subjects								Continuous Assessment-60			End Sem Assessment-40	
									Assignments in classes	Class Performance	Class Attendance	Assignment on the day of grand viva	Viva-voce (Before board of Examiners)
6.	Basic Science	Applied Physics-II Lab	0	0	2	2	1	100	30	20	10	20	20
7.	Engineering Science	Introduction to IT Systems Lab	0	0	4	4	2	100	30	20	10	20	20
8.	Engineering Science	Fundamentals of Electrical & Electronics Engineering Lab	0	0	2	2	1	100					
9.	Engineering Science	Engineering Mechanics Lab	0	0	2	2	1	100	30	20	10	20	20
	AUDIT COURSES-Mandatory non-credit												
10.	Audit	Indian Constitution	2	0	0	2	0	100	30	20	10	20	20
	Total		13	4	10	27	**20**	**1000**					

Syllabus of Mathematics-II

Course Title:	Engineering Mathematics-II
Course Code:	BS102/M-II
Number of Credits:	4(L: 3+T: 1) P: 0
Pre-Requisites:	Knowledge of the basic Co-ordinate Geometry, Statistics & Differential calculus
Total Contact Hours:	60 hrs.

Aim: Mathematics is the backbone of engineering students. The curriculum of mathematics has undergone changes from time to time in accordance with the need of engineering branches. The syllabus has been designed in keeping view the emerging needs of all categories of students. Great emphasis has been given on the application of various contents. This course will develop analytical abilities to make exact calculations and provide a continuing educational base for the students.

Course Objectives: After the completion of the course, the students will be able to
i) apply the knowledge of Cramer's rule and matrix inversion for finding the solutions of Linear Simultaneous Equations.
ii) apply the equations of a straight line, circle, conic section in solving the practical problems.
iii) apply the various techniques of evaluating integration and various methods of finding complete primitive of ordinary differential equations of 1st order and second order for solving engineering problems.
iv) use the concept of partial differentiation in solving physical problems.
v) analyze the Statistics and Probability in a real situation.

Unit-1

Determinants & Matrices **10 Hours**

1.1 Determinant: **4**
1.1.1 Definition and expansion of determinants of order 2 & 3. Minors and cofactors
1.1.2 Elementary properties of Determinants (statements only) & simple problems
1.1.3 Chios Method for 4th order determinant
1.1.4 Solution of linear simultaneous equations (up to 3 unknowns) by Cramer's Rule.

1.2 Matrix:
1.2.1 Definition of Matrix and its order. **6**
1.2.2 Different types of Matrices.(rectangular, square, row, column, upper triangular, lower triangular, diagonal, scalar, identity, null)
1.2.3 Equality of two matrices
1.2.4 Addition, subtraction, multiplication of a matrix by a scalar and multiplication of two matrices
1.2.5 Transpose of a matrix, symmetric & skew symmetric matrices, simple problems
1.2.6 Singular & non-singular matrices, adjoint and inverse of a matrix of order 3

Syllabus of Mathematics-II

Unit-2

Co-ordinate Geometry (only 2-dimension) **13 Hours**

2.1 Coordinate System **3**
2.1.1 Cartesian & Polar Coordinate system & their relations.
2.1.2 Distance between two points, internal & external division of a line segment & simple problems.
2.1.3 Area of a triangle and condition of collinearity.

2.2 Straight Line **4**
2.2.1 Gradient (slope) of a straight line
2.2.2 Equations of straight line in various standard forms (no proof) (slope-intercept form, slope-point form, intercept form, two point form) & simple problems
2.2.3 Angle between two straight lines- conditions of parallelism and perpendicularity & simple problems
2.2.4 Perpendicular distance from a given point to a line, perpendicular distance between two parallel lines

2.3 Circle: **2**
2.3.1 Equation of a circle - centre-radius form, diameter form, simple problems
2.3.2 General equation of a circle and its centre and radius. Simple problems

2.4 Conic Section: **4**
2.4.1 Definition of conic section, vertex, axis, eccentricity, focus, directrix, latus rectum & problem
2.4.2 Standard equations of parabola and ellipse, simple problems

Unit-3

Integral Calculus **15 Hours**

3.1 Indefinite integral **8**
3.1.1 Definition of Integration as inverse process of differentiation.
3.1.2 Rules for integrations (sum, difference, scalar multiple)
3.1.3 Integration of standard functions
3.1.4 Integration by substitution
3.1.5 Integration by parts
3.1.6 Integration by partial fraction

3.2 Definite Integral **7**
3.2.1 Definition of definite integral & simple problems
3.2.2 Properties of definite integral with simple problems.
3.2. Application of definite integral -i) area of bounded region ii) Volume and surface area of solid generated by revolving an area about x and y-axis

Syllabus of Mathematics-II

Unit-4

Ordinary Differential Equation **10 Hours**

4.1 Definition of ordinary differential equation, order & degree. **1**
4.2 Solution of Differential equation of First order and first degree **5**
4.2.1 Separation of Variables
4.2.2 Homogeneous type
4.2.3 Exact type
4.2.4 Linear type

4.3 **Solution of Linear Second order Differential equations with constant coefficients** **4**
4.3.1 Complementary Functions (C.F)
4.3.2 Particular Integral for polynomial function, e^{ax}, $\sin ax$ and $\cos ax$, [$F(-a^2) \neq 0$], $e^{ax}V$ where V is a polynomial function. Simple problem.

Unit-5

Partial Differentiation **3 Hours**

5.1 Definition & meaning of partial derivative.
5.2 Evaluation of partial derivatives.
5.3 Definition & examples of homogeneous functions.
5.3 Euler's theorem (1st order) on Homogeneous functions for 2 variables (without proof). Problems.

Unit-6

Statistics & Probability **9 Hours**

6.1 Statistics **5**
6.1.1 Definition & examples of frequency distribution.
6.1.2 Measure of Central Tendency (mean, median, mode) for ungrouped and grouped frequency distribution.
6.1.3 Measures of dispersion-Standard deviation, Simple problems

6.2 Probability **4**
6.2.1 Definition of random experiment, sample space, event, occurrence of events & types of events (eg. Impossible, Mutually exclusive, Exhaustive, Equally likely)
6.2.2 Classical definition of probability, simple problems

Syllabus of Mathematics-II

Examination Scheme:

A. Semester Examination pattern of 60 marks:

1. Objective questions- 20 marks (1 mark for each question), (At least 5 questions from each **Group**)
2. Subjective questions- 40 marks (At least one question of 10 marks from each **Group**)

- **Group- A** contains Unit-1 & Unit-2 (At least 20 marks); **Group-B** contains Unit-3 (At least 20 marks); **Group-C** contains Unit-4 (At least 20 marks) , **Group-D** contains Unit-5 & Unit-6 (At least 20 marks)

N.B.- Student will answer objective type questions of 20 marks and for subjective questions of 40 marks, taking one question from each **group** of the above four **groups**.

B. For the internal Assessment 40 marks:

1. Class Test Examination/Internal Examination: 20 marks; choose best two out of three Class Test Examinations/ Internal Examinations
2. Class Attendance: 10 marks
3. Viva/ Quiz/Presentation/Assignment/Project/Report etc.: 10 marks

Text Books & Reference –

1. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, New Delhi
2. Engineering Mathematics, Konch & Dey, Bhagabati Publication
3. Numerical Analysis, N. Islam, Academic Press
5. Introduction to Integral Calculus, rohde, Jain, Poddar,Ghosh, Wiley
6. Higher Algebra: Abstract And Linear, SK Mapa, Sarat Book House
7. Analytic Geometry Two & Three Dimensional and Vector Analysis, R. M. Khan, New Central Book Agency
8. Probability & Statistics for Engineers, Richard A. Jonson, Prentice Hall of India
9. An Introduction to Differential Equations, Ghosh, Maity, New Central Books Agency
10. Higher Engineering Mathematics, B. S. Grewal, Khanna Publishers, New Delhi
11. Engineering Mathematics, A. Sarkar, Naba Publication
12. Engineering Mathematics, Konch & Dey, Bhagabati Publication
13. Engineering Mathematics, Babu Ram, Pearson
14. Diploma Engineering Mathematics, B. K. Paul, U.N.Dhar & Sons
15. Engineering Mathematics, V. Sundaram, R. Balasubramanian, K.A. Lakshminarayanan, Vikas Publishing House.
16. Web portal: https://www.ndl.gov.in/homestudy/science
https://ncertbooks.ncert.gov.in/login
https://epathshala.nic.in//
https://webscte.co.in/
https://en.wikipedia.org/wiki/
https://openlibrary.org/
https://www.youtube.com/
http://content.inflibnet.ac.in/
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Sem-II(Theory), Applied Physics –II

Course Code	:	BS104
Course Title	:	Applied Physics –II
Number of Credits	:	3 (L: 2, T: 1,P: 0)
Prerequisites	:	High School Level Physics
Course- Category	:	BS

Course Objectives

Applied Physics aims to give an understanding of this world both by observation and by prediction of the way in which objects behave. Concrete use of physical principles and analysis in various fields of engineering and technology are given prominence in the course content. The course will help the diploma engineers to apply the basic concepts and principles to solve broad- based engineering problems and to understand different technology based applications.

Course Content

Unit -1: Wave motion and its applications

Simple Harmonic Motion (SHM): definition, expression for displacement, velocity, acceleration, time period, frequency etc. study of vibrations of cantilever and determination of its time period, Free, **damped and forced** vibrations with examples.

Wave motion, transverse and longitudinal waves with examples (Sound and light waves) definitions of wave velocity, frequency and wave length and their relationship, equation of a plane progressive wave. Principle of superposition of waves and beat formation.

Acoustics of buildings- reverberation, reverberation time, methods to control reverberation time , noise, coefficient of absorption of sound, Ultrasonic waves – Introduction and properties, engineering and medical applications of ultrasonic.

Unit – 2: Optics

Basic optical laws: reflection and refraction, refractive index, Images and image formation by thin lenses, lens & lens maker's formula, (no deduction) power of lens, magnification simple numerical problems. Total internal reflection, Critical angle and conditions for total internal reflection, applications of total internal reflection in optical fiber:

Optical Instruments; simple and compound microscope, astronomical telescope (refracting, Ray Diagram and formula for magnification). Interference and diffraction of light (Qualitative ideas only).

Unit – 3: Electrostatics

Coulombs law, unit of charge, Electric field, Electric lines of force and their properties, Electric flux, Electric potential and potential difference, Gauss law (statement only) Application of Gauss law to find electric field due to a charged sphere.

Capacitor and its working, types of capacitors, Capacitance and its units. Capacitance of a parallel plate capacitor (formula only), Series and parallel combination of capacitors formula (related numerical problems), dielectric and its effect on capacitance, dielectric break down.

Unit – 4: Current Electricity

Electric Current and its units, Direct and alternating current, resistance and its units, Specific resistance, Conductance, Specific conductance, Series and parallel combination of resistances. Factors affecting resistance of a wire, carbon resistance and colour coding.

Ohm's law, Kirchhoff's laws, Wheatstone bridge, Carrey Foster Bridge and its applications, Concept of terminal potential difference and Electro motive force (EMF)

Heating effect of current, Electric power, electric energy and its units (related numerical problems)

Thermoelectric effect: Seebeck &Peltier effects.

Unit -5: Electromagnetism

Magnetic field and its origin, units

Lorentz force (force on moving charge in magnetic field).Biot- Savart law, Application to **Straight** Conductor & circular loop; concept of magnetic dipole. Force on current carrying conductor, Torque on rectangular coil placed magnetic field concept of electromagnetic induction, Faraday's Laws, Moving coil galvanometer; principle, construction and working, Conversion of a galvanometer into ammeter and voltmeter.

Types of magnetic materials; dia, para and ferromagnetic with their properties.

Unit-6: Semiconductor Physics

Energy bands in solids, Types of materials (insulator, semi-conductor, conductor), intrinsic and extrinsic semiconductors, p-n junction, junction diode and V-I characteristics, Diode as rectifier- half wave and full wave rectifier (Centre taped) & circuit symbol.

Transistor, Block diagram types (pnp and npn)& circuit symbol, transistor as an amplifier CE mode (Circuit diagram and concept).

Photocells, Solar cells and LED working principle and engineering application.

Unit-7: Modern Physics

Bohr's atom model and concept energy levels, ionization and excitation potentials, X-rays, Production (Coolidge tube) continuous and characteristic-X-rays, soft and hard X-rays, and use,

Laser: spontaneous and stimulated emission; Laser light; He-Ne laser elementary characteristics, applications of lasers.

Fiber Optics: Introduction to optical fibers, mechanism of light propagation, applications.

Nanoscience and nanotechnology (Introduction only).

Reference books:

1. Text books of Physics for Class XI & XII (Part I & II); N.C.E.R.T., Delhi.
2. Applied Physics, Vol. I and Vol. II, TTTI Publications, Tata McGraw Hill, Delhi
3. Concepts in Physics by HC verma, Vol.I & II, Bharti Bhawan Ltd. New Delhi
4. Engineering Physics by PV Naik, Pearson Education Pvt. Ltd, New Delhi
5. Modern approach to Applied Physics-I and II, AS Vasudeva, Modern Publisers.
6. A Textbook of Optics, N Subramanyam, Brij Lal,MN Avahanulu, S Chand and Company Ltd.
7. Introduction to Fiber optics, Ajoy Ghatak and K Thyagarajan, Cambridge University Press India Pvt. Ltd, New Delhi.
8. Nanoscience and Nanotechnology, KK Choudhury, narosa Publishing House, Pvt. Ltd. New Delhi.
9. Nanotechnology: Importance and Applications, M.H. Fulekar, IK International Publishing House Pvt. Ltd., New Delhi

Course Title :	Introduction to IT Systems Theory
Course Code	ES 102
Number of Credits :	2 (L: 2, T: 0, P: 0)
Prerequisites	NIL
Course Category	ES
Course code : General	Semester : SECOND
Duration : 16 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 2 hrs/week	Continuous Internal Assessment : 20 Marks
Tutorial: - 1 hr/week	Attendance, Assignment & Quiz : - 20 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	Develop basic concept of Computer Science

Course Objectives:: This course is intended to make new students comfortable with computing environment - Learning basic computer skills, Learning basic application software tools, Understanding Computer Hardware, Cyber security awareness.

Course Content:

Contents (Theory)	Hrs./Unit	Marks
UNIT 1:	10	28

Basic Internet skills: Understanding browser, efficient use of search engines, awareness about Digital India portals (state and national portals) and college portals.

Number system and codes: Binary, octal, hexadecimal and decimal Number systems and their inter conversion, BCD numbers (8421-2421), gray code, excess–3 code, cyclic code, code conversion, ASCII, EBCDIC codes. Binary addition and subtraction, signed and unsigned binary numbers, 1's and 2's complement representation. 2. Boolean Algebra: Basic logic circuits: Logic gates (AND, OR, NOT, NAND, NOR, Ex-OR, ExNOR and their truth tables,), Universal Gates, Laws of Boolean algebra, De-Morgan's theorem

General understanding of various computer hardware components – CPU, Memory(types), Display(modern), Keyboard, Mouse, HDD, SSD and

other Peripheral Devices.
Types of software

UNIT 2:	**10**	28

Overview of Operating Systems

- **What is an OS**
- **Brief history.**

Background and Basics

- **Computer System review**
- **Types of OS**
- **Architecture**
- **Basic Oss**
- **Batch**
- **Multi-programmed batch**
- **Timesharing**
- **Computer System Structures**
- **Operating System Structures**

UNIT 3:	**2**	8

Algorithm and Flowcharts

- **Algorithm**
- **Definition**
- **Characteristics**

Advantages and disadvantages

- **Flowchart**
- **Definition**
- **Define symbols of flowchart**
- **Advantages and disadvantages**

Examples

UNIT 3:	**7**	20

HTML 5: • HTML – Introduction • HTML – Elements • HTML – Tags • HTML – Text • HTML – Formatting • HTML – Pre • HTML – Attributes • HTML – Font • HTML – Text Links • HTML – Comments • HTML – Lists • HTML – Images • HTML – Image Links • HTML – Tables • HTML – Bgcolor • HTML – Color Codes • HTML – Color Chart • HTML – Background • Web Forms • HTML – Forms • HTML – Input • HTML – Text Fields • Hidden Fields • HTML – Password • HTML – Reset • HTML – Submit • HTML – Checkboxes • HTML – Radio • HTML – Select • HTML – Hidden Fields • HTML – Upload • HTML – Textarea • Special Tags • HTML – Body • HTML – Meta • HTML – Style • HTML – Div • HTML – Layouts • HTML – Frames • Formatting Tags • HTML – Bold • HTML – Paragraphs • HTML – Headings • HTML – Line Breaks

CSS: CSS Introduction • CSS Syntax • CSS Id & Class • CSS How CSS Styling • Styling Backgrounds • Styling Text • Styling Fonts • Styling Links • Styling Lists • Styling Tables CSS Box Model • CSS Border • CSS Outline • CSS Margin• CSS Display • CSS Positioning • CSS Floating • CSS Navigation Bar • CSS Image Gallery • CSS Image Opacity • CSS Align **JavaScript: Introduction • JavaScript Overview • JavaScript Syntax • Type of JavaScript • Embedding Script In HTML File • Variable Operators • Arithmetic • Logical • Comparison • Assignment • Conditional Conditional Statement & Looping Statement • If • If. Else • Switch • While • Do/while**		
UNIT 5:	**3**	16
(Network Utilities and devices tools/project) **1: Introduction to Computer Security Chapter** **2: Networks and Internet (DoS Tools & Techniques – Tracert, Visual Route)** **3: Cyber Stalking, Fraud, and Abuse** **4: Denial of Service Attacks (Scanning – WireShark)** **5: Malware Chapter 6: Techniques Used by Hackers.**		
Reference Book		
1. Computer Fundamentals by Goel, Pearson; **2. Computer Architecture and Maintenance. By - Dr. Sachin Publisher - Shroff Publisher;** **3. Introduction to Computer Science, ITL Education Solutions Limited, Pearson.** **4. FUNDAMENTALS OF COMPUTERS by E Balagurusamy. McGraw Hill Education;** **5. Express Learning - Computer Fundamentals and Programming, By Ashok Kamthane/ITL ESL. Pearson;**		
Course outcomes: **At the end of the course student will be able to comfortably work on computer, install and configure OS, assemble a PC and connect it to external devices, basic design of web page, protect information and computers from basic abuses/attacks.**		

Fundamental of Electrical & Electronics Engineering (FEEE)

Sl. No.	Code No.	Course Title	Hours per week			Semester	Credits
1.	ES 104	FUNDAMENTAL OF ELECTRICAL & ELECTRONICS ENGINEERING (FEEE)	2	1	0	II	3
2.	ES 110	FUNDAMENTAL OF ELECTRICAL & ELECTRONICS ENGINEERING (FEEE) (Laboratory)	0	0	2	II	1

Course Outcomes:

After completion of this course students will be able to

- Realize the concept and use of different Electrical and Electronics components.
- Understand the basics of electrical and electronics engineering principles.
- Understand the basics of AC circuits, Transformers and Machines.
- Realize the fundamental principle of Semiconductor, P-N junction Diode and Transistor.
- Know the basic working principle of OPAMP.
- Understand the functions of basic Digital Logic Gates.

Units	Theory Content (Name of topic)	Total nos. of period/ hours required
	Overview of Electrical Components	3
Unit 1	1.1 Concept of Passive Components: Resistors, Capacitors, Inductors. 1.2 Different types of signal waveforms: DC and AC, periodic and non-periodic. 1.3 Concept of Ideal and practical Voltage and current sources and their transformation.	
	Electric and Magnetic Circuits	7
Unit 2	2.1 Concept of EMF, Current, Potential Difference, Power and Energy 2.2 M.M.F, magnetic force, permeability, hysteresis loop, reluctance, leakage factor, BH curve and hysteresis loss, Eddy current loss. Analogy between electric and magnetic circuits 2.3 Electromagnetic induction: Faraday's laws, Lenz's law Dynamically induced emf; Statically induced emf; Principle of self and mutual inductance 2.4 Flemming's Left Hand Rule and Right Hand Rule	
	A.C. Circuits	8
Unit 3	3.1 Concept of Angular speed, Cycle, Frequency, Periodic time, Amplitude, RMS value, Average value, Form Factor, Peak Factor, impedance, phase and phase difference. 3.2 Expressions of voltage and current through pure resistance, inductance and capacitance with sinusoidal Excitation and phasor representation. 3.3 Study of simple R-L, R-C, R-L-C series and parallel circuits with sinusoidal excitation and phasor representation.	

	3.4 Concept of impedance, impedance triangle, power factor, active, reactive and apparent power, power triangle. 3.5 Voltage and Current relationship in Star and Delta connections (no deductions) and simple problems.	
	Transformer and Machines	6
Unit 4	4.1 Basic working principle of two winding transformers 4.2 EMF equation and transformation ratio of transformers 4.3 Basic working principle and applications of AC and DC motors 4.4 Simple problems on transformer and electrical motors	
	Overview of Basic Semiconductor Devices	6
Unit 5	5.1 Energy level diagram of Insulator, Conductor and Semiconductor, Concept of Intrinsic and Extrinsic semiconductor 5.2 Formation of P-Type and N-Type semiconductor and their properties. 5.3 Formation of P-N junction Diode, Forward and Reversed biased characteristics. 5.4 Concept of PNP and NPN Transistor with their applications in CE mode.	
	Overview of Analog Circuits	5
Unit 6	6.1 Features of an ideal OPAMP, Pin configuration of 741C, Concept of Virtual Ground 6.2 Inverting and non-inverting OPAMP amplifiers and their gain calculation. 6.3Applications of OPAMP Amplifiers, adder	
	Overview of Digital Electronics	5
Unit 7	7.1. Logic gates (AND, OR, NOT, NAND, NOR, Ex-OR, Ex-NOR and their truth tables), Universal Logic Gates. 7.2 Postulates of Boolean Algebra 7.3 De-Morgan's theorem, Minterm (SOP), Maxterm (POS)	
	Total Periods/Hours	**40**

	Practical Outcome	**Total nos. of period/ hours required**
1.	Measure voltage, current and power in a 1-phase circuit with resistive load.	2
2.	Visualize phase difference between voltage and current in series R-Land R-C circuits with the help of oscilloscope and plot the phasor diagram.	2
3.	Measure voltage, current, power and power factor in a R-L series circuit.	2
4.	Identify different parts of a single-phase transformer, A.C./D.C. Motor.	2
5.	Determine the transformation ratio of a single-phase transformer and measure no load current of it.	2
6.	Identify various active and passive electronic components in a given circuit.	2
7.	Familiarization with multimeter to measure the value of given resistor and to verify with colour code.	1
8	Use of LCR-Q meter to measure the value of a given capacitor and inductor.	2
9.	Test the PN-junction diodes using digital multimeter and find out their V-I characteristics in forward biased circuit.	2
10.	Identify three terminals of a transistor by using digital multimeter	2
11.	Use of Op-Amp as Inverting and non-inverting amplifier.	2
12.	Realization of Truth Table of different Logic Gates and verification of De-	2

	Morgan's theorem.	
	Total Periods/ Hours	**23**

• List of Recommended Books:

(1). Basic Electricals and Electronics--------- S.K. Bhattacharya (Pearson Education)
(2). Fundamental of Electrical and Electronics Engineering------ Samarjit Ghosh (PHI)
(3). Basic Electrical and Electronics Engineering------ J.B. Gupta (S.K. Kataria & Sons)
(4). Basic Electrical and Electronics Engineering------- R. Math Subramanian & S. Salivahanan (McGraw Hill Education)
(5). Principle of Electrical Engineering and Electronics-------- V. K. Mehta & Rohit Mehta (S.Chand)

Course Title: Engineering Mechanics
Number of Credits: 3 (L: 2, T: 1, P: 0)
Course Category: ES
Course Code: ES 106

Course Objectives:

Followings are the objectives of this course:

1. To obtain resultant of various forces
2. To understand basic laws of motion and their applications
3. To understand role of friction in equilibrium problems
4. To know fundamental laws of machines and their applications to various engineering problems
5. To know fundamental laws of motions and their applications to various engineering problems

Course Contents:

Group A

Unit – I Basics of Mechanics and Force System [4L+2T]

Concept of Engineering Mechanics – Statics & Dynamics; Space, time, mass, particle, flexible body and rigid body. Scalar Quality and Vector Quality; Addition & Subtraction of Vectors – Basic units – Derived Units – SI units.

Force: Definition, unit, representation as a vector and by Bow's notation, characteristics and effects of a force, Principle of transmissibility of force. Force systems and its classification with special emphasis of Co-planar Force System.

Co-planar Con-current Force System: Composition of Forces: Parallelogram Law, Triangle Law and Polygon Law of Forces. Determination of resultant by Analytical and Graphical method. Vector diagram. Resolution of Forces: Orthogonal components of a force. Simple problems on composition and resolution of forces

Unit – II Moments and Couples [4L+2T]

Moment: Definition of moment of a force about a point – Physical significance of moment – Resultant of a system of parallel and inclined forces – Varignon's Theorem. Simple Problems

Couples: Definition of moment of a couple – Physical significance of Couples Equivalent couples – Resultant of any number of coplanar couples – Replacement of a force about a point by an equal like parallel force together with a couple. Simple Problems

Unit – III Condition of Equilibrium [5L+3T]

Equilibrium of Co-planar Con-current Force System: Lami's Theorem. Triangle Law & Polygon Law of equilibrium Concept of Free Body diagram. Simple problems

Equilibrium of Co-planar system of non-concurrent forces: Conditions of equilibrium of co-planar system of non-concurrent parallel forces (Like and Unlike) Simple problems excluding statically in-determinant

Types of beams, supports (simple, hinged, roller and fixed) and loads. Beam reaction for simply supported beam with or without overhang subjected to combination of Point Load and Uniformly Distributed load. Simple Problems

Unit – IV Friction **[4L+3T]**

Friction and its relevance in engineering, Types and Laws of friction, Limiting Friction, Co-efficient of friction, Angle of friction, Cone of Friction, Angle of Repose, Relation between co-efficient of friction and angle of friction.

Equilibrium of bodies on horizontal surface subjected to force parallel and inclined to plane.

Equilibrium of bodies on inclined plane subjected to force parallel and inclined to plane. Simple Problems

Group B

Unit– V Centroid and Centre of Gravity **[4L+2T]**

Concept & definition.

Determination of Centroid of Plane Figures like (i) Uniform triangular lamina, (ii) Uniform rectangular lamina, (iii) Uniform circular lamina, (iv) Uniform semi-circular lamina, and, (v) Uniform lamina of quadrant of a circle.

Determination of Centroid of Composite sections composed of not more than three plane figures like (i) T-section, (ii) equal and unequal angle-sections, (iii) equal and unequal I-sections, (iv) Channel-sections, (v) Z-sections, (vi) different cut-out sections, and, (vii) Built-up sections (Data of individual sections given). Simple Problems

Concept of Centre of Gravity of Simple Solids. Determination of Centre of Gravity of Simple Solids like (i) Cube (ii) Cuboid (iii) Cylinder (iv) Sphere. Determination of Centre of Gravity of composite solids composed of not more than two simple solids. No Problem

Unit– VI Simple Lifting Machines **[5L+2T]**

Definition of Lifting Machine, Applications and Advantages.

Load, Effort, Mechanical Advantage, Velocity Ratio, Efficiency of Machines and their relationships. Law of machine.

Ideal Machine, Friction in Machine, Maximum Mechanical Advantage and Efficiency, Reversible and non-reversible machines, conditions for reversibility. Simple Problems

Velocity ratios of (i) Simple and Differential Axle and Wheel (ii) Worm and worm wheel (iii) Single purchase and Double Purchase Crab Winch (v) Simple Screw Jack (vi) Simple Pulley Block. Simple numerical problems

Group C

Unit– VI Motion in a Plane [4L+2T]

Rectilinear Motion

Displacement- Time and Velocity-Time diagrams, Motion equations (with deduction). Newton's Second Law of linear motion P = ma and momentum of a body. Conservation of momentum of a body. Simple numerical problems

Curvilinear Motion

Concept and definition of Angular displacement, Angular velocity, relation between Linear & Angular velocity. Definition of Angular acceleration, Relation between linear & angular acceleration Concept, definition and deduction of expression for Centripetal and centrifugal force (numerical problems

Work, Power, Energy

Concept, definition and mathematical expression of Work, Power and Energy. Discussion on their units in SI System.

Suggested Learning Resources:

1. S.S. Bhavikatti, Engineering Mechanics, New Age International (P) Ltd., Publishers
2. D.S. Bedi, Engineering Mechanics, Khanna Publications, New Delhi (2008)
3. Bansal R K, A Text Book of Engineering Mechanics, Laxmi Publications
4. Ramamrutham, Engineering Mechanics, S. Chand & Co. New Delhi.
5. A.R.Basu, A Text Book of Engineering Mechanics
6. Khurmi, R.S., Applied Mechanics, S. Chand & Co. New Delhi
7. D.P. Sharma, Engineering Mechanics, Pearson Publications
8. S. Timosenko and D.H. Young, Engineering Mechanics, McGRAW-HILL Book Company
9. Meriam, J. L., Kraige, L.G., Engineering Mechanics, Wiley Publication, New Delhi

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.**Course outcomes:**

After completing this course, student will be able to:

1. Identify the force systems for given conditions by applying the basics of mechanics.
2. Determine unknown force(s) of different engineering systems.
3. Apply the principles of friction in various conditions for useful purposes.
4. Find the centroid and centre of gravity of various components in engineering systems.
5. Select the relevant simple lifting machine(s) for given purposes
6. Apply the basic laws of motion in engineering practices

Course Title : Engineering Mechanics Lab.
Number of Credits: 1 (L: 0, T: 0, P: 2)
Course Category: ES
Course Code: ES 112

Course Objectives::

Followings are the objectives of this course:

1. To obtain resultant of various forces
2. To calculate support reactions through conditions of equilibrium for various structures
3. To understand role of friction in equilibrium problems
4. To know fundamental laws of machines and their applications to various engineering problems

List of Practicals to be performed:

[Total Six Practicals out of which any three from Sl. No. 1to 5 and any three from the rest]

1. To find the M.A., V.R., Efficiency and law of machine using Single purchase crab winch
2. To find the M.A., V.R., Efficiency and law of machine using Double purchase crab winch.
3. To find the M.A., V.R., Efficiency and law of machine using Worm and worm wheel.
4. To find the M.A., V.R., Efficiency and law of machine for Differential Axle and Wheel.
5. To find the M.A., V.R., Efficiency and law of machine for Simple Screw Jack.
6. To determine coefficient of friction for motion on horizontal plane.
7. To determine coefficient of friction for motion on inclined plane.
8. To determine resultant of concurrent force system applying Law of Polygon of forces
9. To determine resultant of concurrent force system graphically.
10. To determine resultant of concurrent force system graphically.
11. To determine resultant of parallel force system graphically.
12. To verify Lami's theorem graphically
13. To determine centroid of geometrical plane figures
14. Study forces in various members of Jib crane.

Course outcomes:

After completing this course, student will be able to

1. Identify the force systems for given conditions by applying the basics of mechanics.
2. Determine unknown force(s) of different engineering systems.
3. Apply the principles of friction in various conditions for useful purposes.
4. Find the centroid and centre of gravity of various components in engineering systems.
5. Select the relevant Simple Lifting Machine(s) for given purposes

Examination Scheme:

A. Semester Examination : 60 marks

1. Objective questions: 20 marks (1 mark for each question)
   Questions should be set in a ratio of 5:3:2 from Group A, B and C.
   Objective questions are compulsory for the students

2. Subjective questions:40 marks
   10 questions [5 questions from Group A, 3 from Group B and 2 from Group C] each of 8 marks are to set.
   Students will answer 5 (five) questions taking at least 2 from Group A, 1 from Group B and 1 from Group C.

B. Internal Assessment: 40 marks:

1. Class Test Examination/Internal Examination: 20 marks
   Best two out of three Class Test Examinations/Internal Examinations each of 20 marks

2. ClassAttendance:10marks

3. Viva/ Quiz/Presentation/Assignment/Project/Report etc.:10 marks

SEM-II(LAB), Applied Physics II Lab

Course Code	:	BS106
Course Tittle	:	Applied Physics II Lab
Number of Credits	:	1 (L:0, T:0, P:2)
Prerequisites	:	NIL
Course Category	:	BS

Course Objectives:

Concrete use of physical principles and analysis in various fields of engineering and technology is very prominence. The course aims to supplement the factual knowledge gained in the lecture by first hand manipulation of apparatus. This will develop scientific temper and help to apply the basic concepts and principles in solving engineering and technology based problems. In addition, students get necessary confidence in handling equipment and thus learn various skills in measurement.

List of Practicals/Activites: (To perform minimum 8 Practicals)

1. To determine and verify the time period of **oscillation of** a cantilever.
2. To verify laws of refraction (Snell's law) using a glass slab.
3. To determine focal length and magnifying power of a convex lens by **u-v method**.
4. To verify Ohm's law by plotting graph between current and potential difference.

5.a.To verify laws of resistances in series by P.O.box.

5.b. To verify laws of resistances in parallel by using Ammeter and Volt meter.

6.To verify Kirchhoff's law using electrical circuits.

7.To find resistance of a galvanometer by half deflection method.

8.To convert a galvanometer into an ammeter.

9.To convert a galvanometer into a voltmeter.

10.To verify inverse square law of radiations using a photo-electric cell.

11. To draw V-I characteristics of a semiconductor diode (Ge, Si) and determine its knee voltage.
12. To study the dependence of capacitance of a parallel plate capacitor on various factors and determine **the** permittivity of air at a place.

Recommended Books:

1. Text Books of Physics for Class XI& XII (Part-I, Part-II); N.C.E.R.T., Delhi
2. Comprehensive Practical Physics, Vol-I & II, JN Jaiswal, Laxmi Publication(P)Ltd., New Delhi
3. Practical Physics by C.L. Arora, S. Chand & Company Ltd.

Course Title : Introduction to IT Systems Lab	
Course Code	ES 108
Number of Credits :	2 (L:, T: 0, P: 3)
Prerequisites	NIL
Course Category	ES
Course code : General	Semester : SECOND
Duration : 16 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
	Continuous Internal Assessment: 40 Marks
	Attendance, Assignment & Quiz : - 20 Marks
Practical : 4 hrs/week	End Semester Examination: 40 Marks
Aim:	Develop basic concept of Computer Science
1 Browser features, browsing, using various search engines, writing search queries	
2 Visit various e-governance/Digital India portals, understand their features, services offered	
3 COMPUTER FUNDAMENDALS Computer and operating system-fundamentals of computer-components of computer system-Input and Output Devices-Memory handling-Storage devices	
4 Read Wikipedia pages on computer hardware components, look at those components in lab, identify them, recognise various ports/interfaces and related cables, etc.	
5 Install Linux and Windows operating system on identified lab machines, explore various options, do it multiple times	
6 Connect various peripherals (printer, scanner, etc.) to computer, explore various features of peripheral and their device driver software.	
7 Practice HTML commands with CSS try them with various values, make your basic own Webpage	
8 MS Excel	
Apply Custom Formats and Layouts	A Learner is able to apply skills in Advanced Excel, is able to – • Format Cells, • Apply Custom Values and predefined Formats • Apply Borders, • Design Borders • Custom Formatting
Create advanced formulas	A Learner is able to Use Simple and Advanced Formulas like – • Nested if, • Reference formulas like –

	• lookup, vlookup, hlookup, • count formula with conditions • Index, Match, • Conditional Loops, etc
Use Scenarios	A Learner is able to seek use Goal Seek function, alter scenarios and values in a cell to reach a goal.
Create Advanced Charts	A Learner can tell where to use what type of charts, and obtain graphical Charts in various scenarios 3D-Graphs, Bar Charts, Pie Chart, Histograms, Line Graph, Sparklines, trend, etc.
Pivot tables & charts	A Learner is able to Apply • Pivot Tables, • Design Pivot Table, • Customize Values,
Manage and Share Workbooks	A Learner is able to Share Workbook Online, email, save on cloud, edit it Online in Google Sheets, Add Collaborators etc.
9 MS PowerPoint	
Create a Power Point presentation using slide template. **Create a Power Point presentation using animation.** **Create a Power Point presentation using transition** **Create a Power Point Presentation with Adding movie and sound.** **Create a Power Point Presentation with Adding tables and chart etc.** **Changing slide colour scheme in presentation.** **Viewing the presentation using slide navigator.** **Create, Save, Run and Print the Power Point Presentation.**	
10	
Create and share files/folders in Google drive **Create and share Google docs.** **Create and share Google sheets.** **Create and share Google Forms.** **Create and share Google slides.**	
Course outcomes: **At the end of the course student will be able to comfortably work on computer, install and configure OS, assemble a PC and connect it to external devices, write documents, create worksheets, prepare presentations, web pages, protect information and computers from basic abuses/attacks.**	
Reference Book	
1. My Office 2016, Pearson **2. Head First Excel : A Brain-Friendly Guide. Publisher: Shroff/O'Reilly** **3. My Excel 2016, Pearson**	

INDIAN CONSTITUTION

Course Code	AU302
Course Title	Indian Constitution
Number of Credits and L-T-P	0 [L – 2, T – 0, P - 0]
Prerequisites	None
Course Category	AU

Course Content

Unit 1 – The Constitution – Introduction

- The History of the Making of the Indian Constitution
- Preamble and the Basic Structure, and its interpretation
- Fundamental Rights and Duties and their interpretation
- State Policy Principles

Unit 2 – Union Government

- Structure of the Indian Union
- President – Role and Power
- Prime Minister and Council of Ministers
- Lok Sabha and Rajya Sabha

Unit 3 – State Government

- Governor – Role and Power
- Chief Minister and Council of Ministers
- State Secretariat

Unit 4 – Local Administration

- District Administration
- Municipal Corporation
- Zila Panchayat

Unit 5 – Election Commission

- Role and Functioning
- Chief Election Commissioner
- State Election Commission

Suggested Learning Resources:

Sl. No.	Title of Book	Author	Publication
1	Ethics and Politics of the Indian Constitution	Rajeev Bhargava	Oxford University Press, New Delhi, 2008
2	The Constitution of India	B.L. Fadia	Sahitya Bhawan; New edition (2017)
3	Introduction to the Constitution of India	DD Basu	Lexis Nexis; Twenty-Third 2018 edition

Suggested Software/Learning Websites:

a. https://www.constitution.org/cons/india/const.html
b. http://www.legislative.gov.in/constitution-of-india
c. https://www.sci.gov.in/constitution
d. https://www.toppr.com/guides/civics/the-indian-constitution/the-constitution-of-india/